Telemedicine App- OptumCure Documentation

Tech Stack

- **Backend-** NodeJS, ExpressJS, MySQL.
- **Frontend-** HTML, CSS, Bootstrap-5, Javascript.
- **Database-** MySQL Database.
- **Deployment-** Azure App Service to deploy Node app, and Azure Relational Database Services to deploy MySQL database.

Requirements (to run on local)

- Xampp- To setup local servers: Apache and MySQL.
- PhpMyAdmin- To handle and manage MySQL database.
- Node- Version 16
- Other Dependencies- Run "npm install" in root directory of the project after cloning from GitHub (described below).

How to setup and run locally

1. **Clone Git Repository** – Run "git clone https://github.optum.com/rjoshi38/telemedicine.git" wherever want to make directory of the project.
2. **Install Dependencies**- Run "npm install" in the root directory after cloning the repository.
3. **Setup Database-** Following is the schema with all the details to construct a MySQL database in phpMyAdmin.

Database-Schema

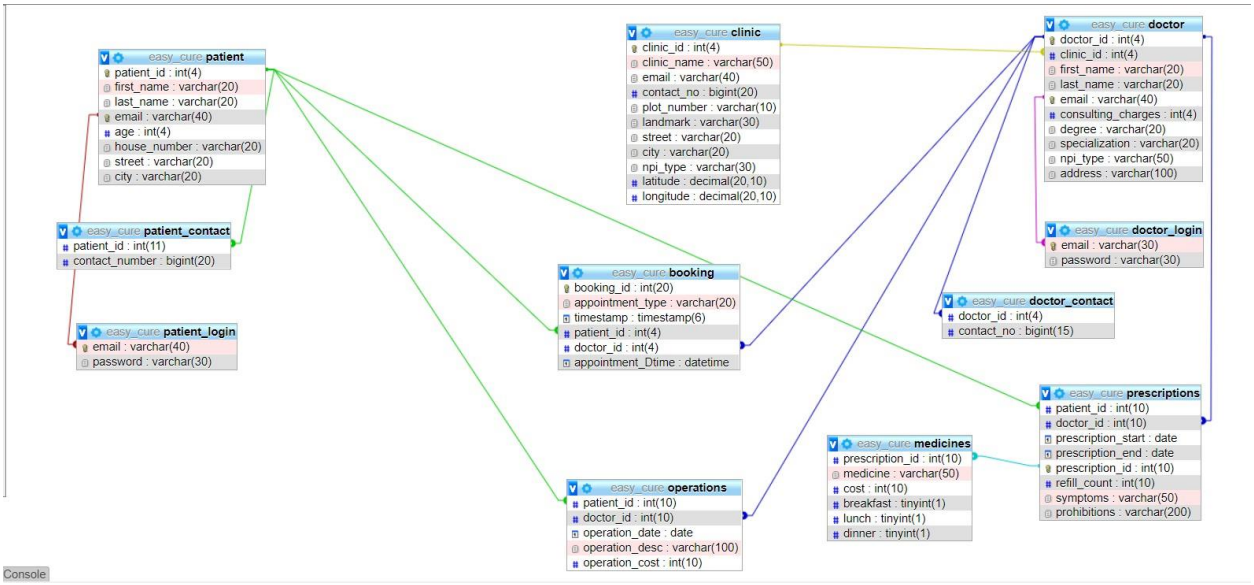
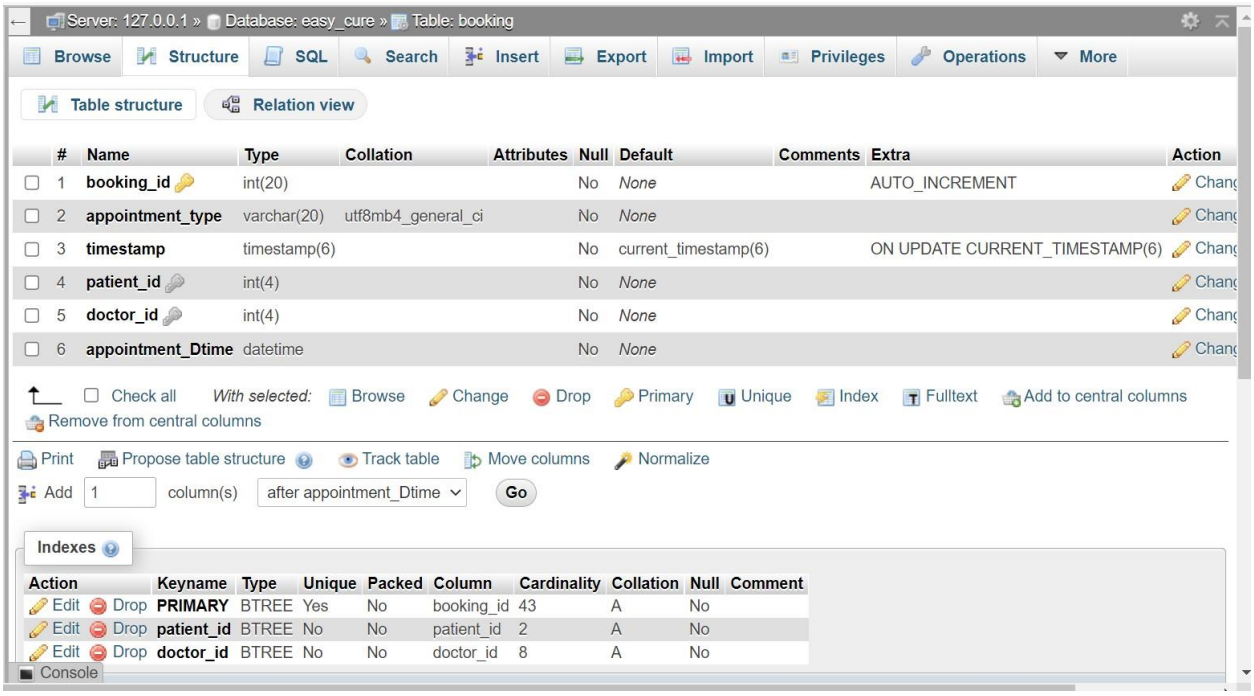


Table-booking



Server: 127.0.0.1 » Database: easy_cure » Table: booking

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | More

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	**booking_id**	int(20)			No	*None*		AUTO_INCREMENT	Chang
2	**appointment_type**	varchar(20)	utf8mb4_general_ci		No	*None*			Chang
3	**timestamp**	timestamp(6)			No	current_timestamp(6)		ON UPDATE CURRENT_TIMESTAMP(6)	Chang
4	**patient_id**	int(4)			No	*None*			Chang
5	**doctor_id**	int(4)			No	*None*			Chang
6	**appointment_Dtime**	datetime			No	*None*			Chang

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after appointment_Dtime Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	**PRIMARY**	BTREE	Yes	No	booking_id	43	A	No	
Edit Drop	**patient_id**	BTREE	No	No	patient_id	2	A	No	
Edit Drop	**doctor_id**	BTREE	No	No	doctor_id	8	A	No	

Console

Table-clinic

Server: 127.0.0.1 » Database: easy_cure » Table: clinic

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	clinic_id	int(4)			No	None		AUTO_INCREMENT	Change Drop More
2	clinic_name	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
3	email	varchar(40)	utf8mb4_general_ci		No	None			Change Drop More
4	contact_no	bigint(20)			No	None			Change Drop More
5	plot_number	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
6	landmark	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
7	street	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
8	city	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
9	npi_type	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
10	latitude	decimal(20,10)			No	None			Change Drop More
11	longitude	decimal(20,10)			No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after longitude Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	clinic_id	3	A	No	

Console

Table-doctor

Server: 127.0.0.1 » Database: easy_cure » Table: doctor

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	doctor_id	int(4)			No	None		AUTO_INCREMENT	Change Drop More
2	clinic_id	int(4)			No	None			Change Drop More
3	first_name	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
4	last_name	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
5	email	varchar(40)	utf8mb4_general_ci		No	None			Change Drop More
6	consulting_charges	int(4)			No	None			Change Drop More
7	degree	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
8	specialization	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
9	npi_type	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
10	address	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after address Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	doctor_id	9	A	No	
Edit Drop	email	BTREE	Yes	No	email	9	A	No	
Edit Drop	clinic_id	BTREE	No	No	clinic_id	9	A	No	

Console

Table-doctor_contact

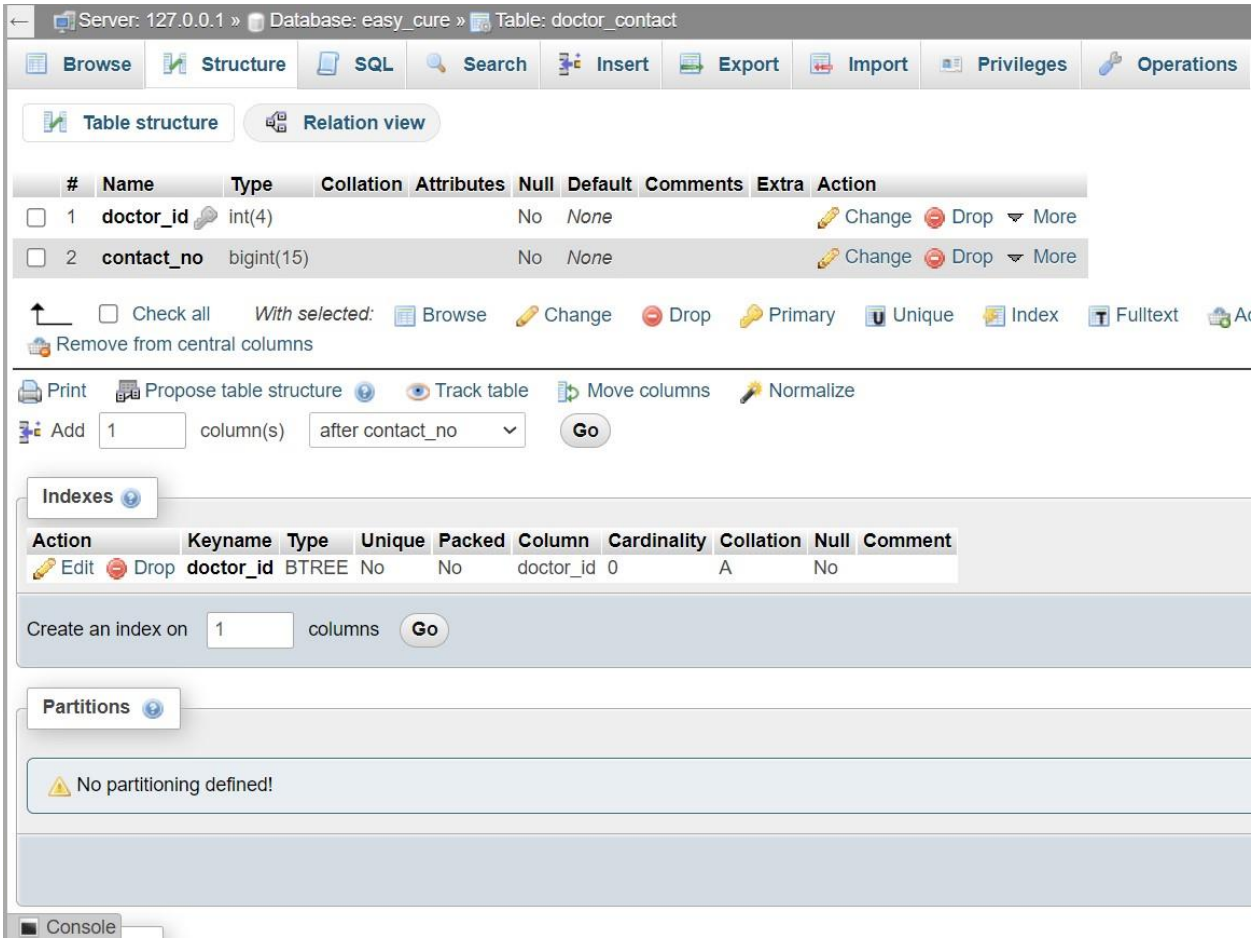


Table-doctor_login

Server: 127.0.0.1 » Database: easy_cure » Table: doctor_login

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	email	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More
2	password	varchar(30)	utf8mb4_general_ci		No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after password Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	email	0	A	No	
Edit Drop	email_2	BTREE	Yes	No	email	0	A	No	
Edit Drop	email	BTREE	No	No	email	0	A	No	

Create an index on 1 columns Go

Partitions

Table-medicines

Server: 127.0.0.1 » Database: easy_cure » Table: medicines

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	prescription_id	int(10)			No	None			Change Drop More
2	medicine	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
3	cost	int(10)			No	None			Change Drop More
4	breakfast	tinyint(1)			No	None			Change Drop More
5	lunch	tinyint(1)			No	None			Change Drop More
6	dinner	tinyint(1)			No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after dinner Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	prescription_id	BTREE	No	No	prescription_id	8	A	No	

Create an index on 1 columns Go

Table-operations

Server: 127.0.0.1 » Database: easy_cure » Table: operations

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	patient_id	int(10)			No	None			Change Drop More
2	doctor_id	int(10)			No	None			Change Drop More
3	operation_date	date			No	None			Change Drop More
4	operation_desc	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
5	operation_cost	int(10)			No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after operation_cost Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	patient_id	BTREE	No	No	patient_id	0	A	No	
Edit Drop	doctor_id	BTREE	No	No	doctor_id	0	A	No	

Create an index on 1 columns Go

Partitions

Table-patient

Server: 127.0.0.1 » Database: easy_cure » Table: patient

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

Table structure | Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	patient_id	int(4)			No	None		AUTO_INCREMENT	Change Drop More
2	first_name	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
3	last_name	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
4	email	varchar(40)	utf8mb4_general_ci		No	None			Change Drop More
5	age	int(4)			No	None			Change Drop More
6	house_number	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
7	street	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
8	city	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Check all *With selected:* Browse Change Drop Primary Unique Index Fulltext Add to central columns Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after city Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Drop	PRIMARY	BTREE	Yes	No	patient_id	2	A	No	
Edit Drop	email	BTREE	Yes	No	email	2	A	No	
Edit Drop	patient_id	BTREE	No	No	patient_id	2	A	No	

Table-patient_contact

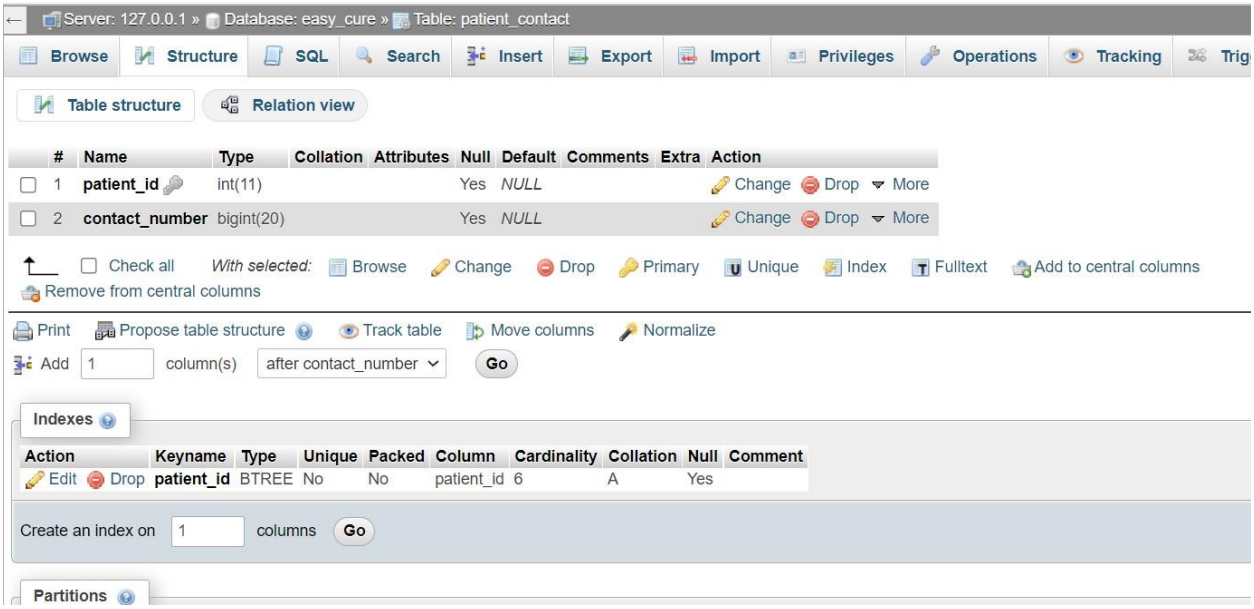


Table-patient_login

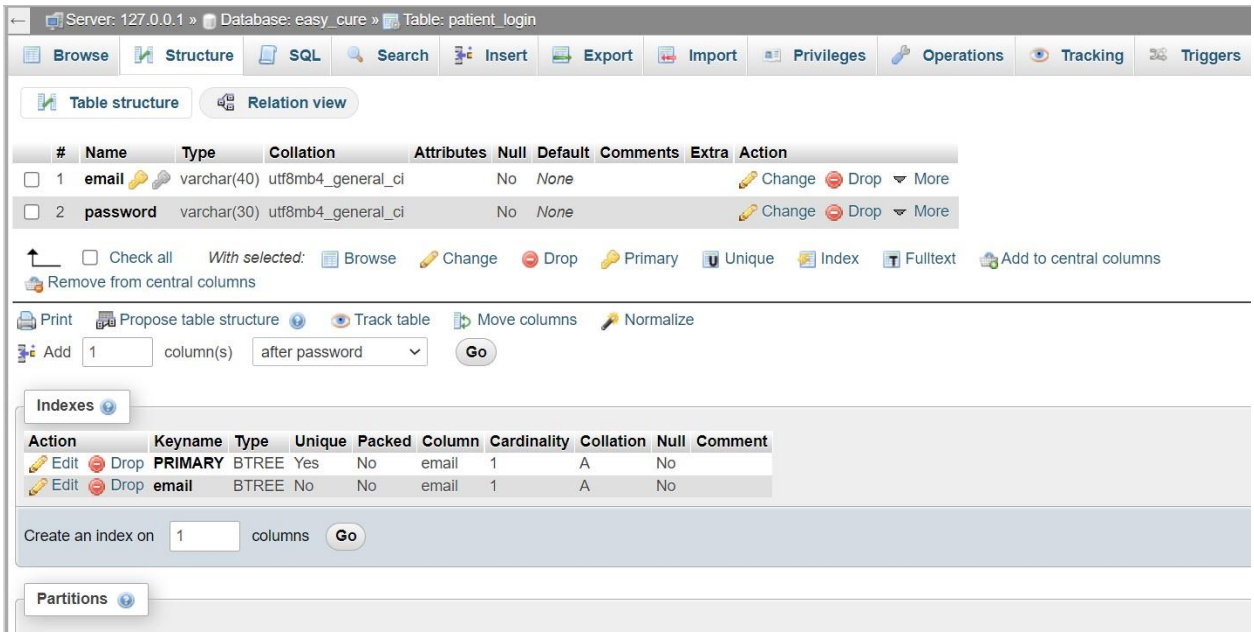
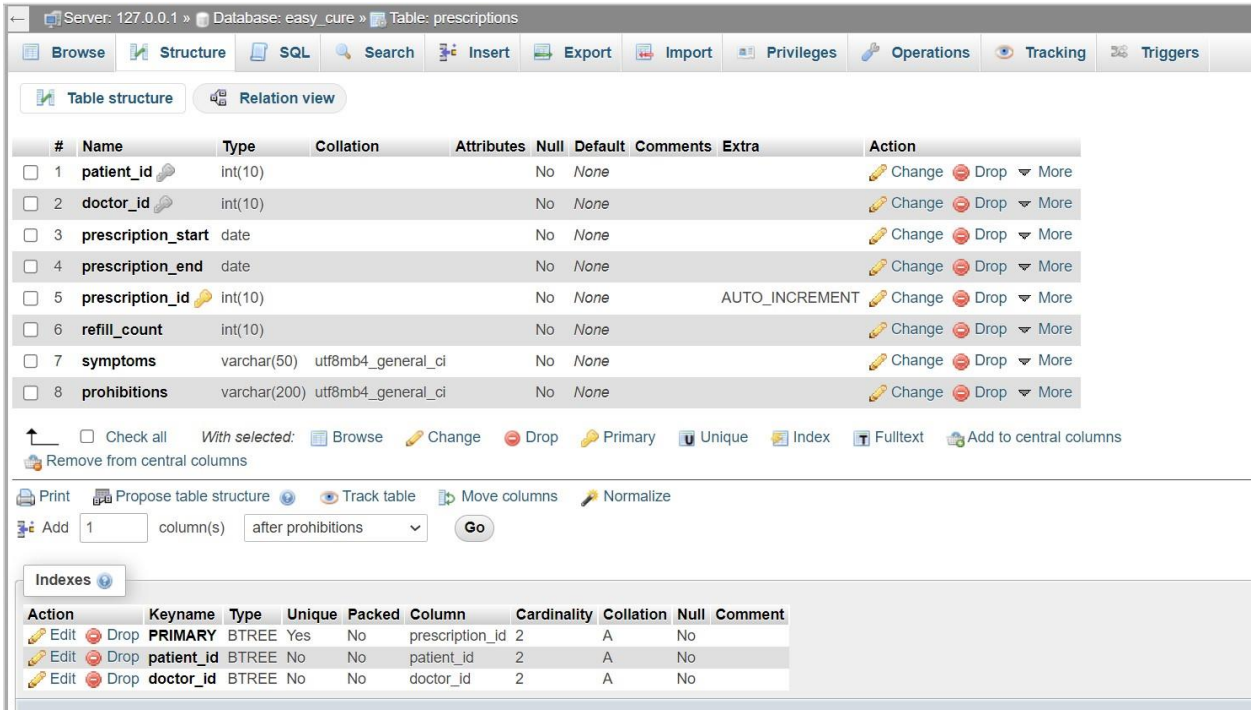


Table-prescription



4. **Start Command-** After installing dependencies and setting up the database run “node app” at the terminal in the root directory of the project.
5. Project is ready to use and test locally now and can be accessed on the port number stated in the app.js file via any browser.

GMAIL AND GOOGLE CALENDAR INTEGRATION

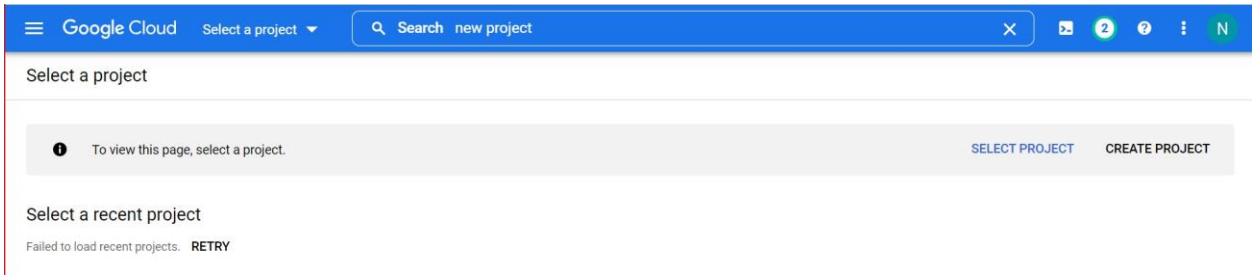
Here we are using GOOGLE CALENDAR API for the google calendar integration and the Gmail integration.

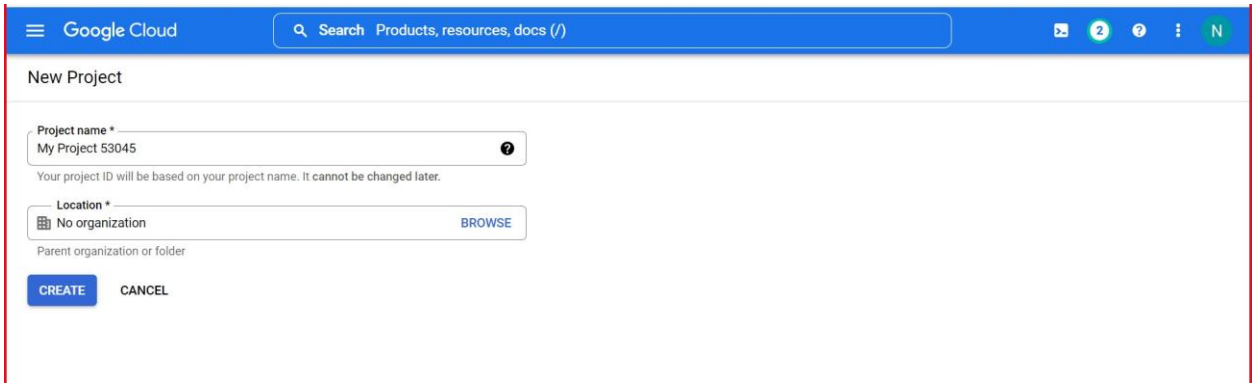
STEP 1:

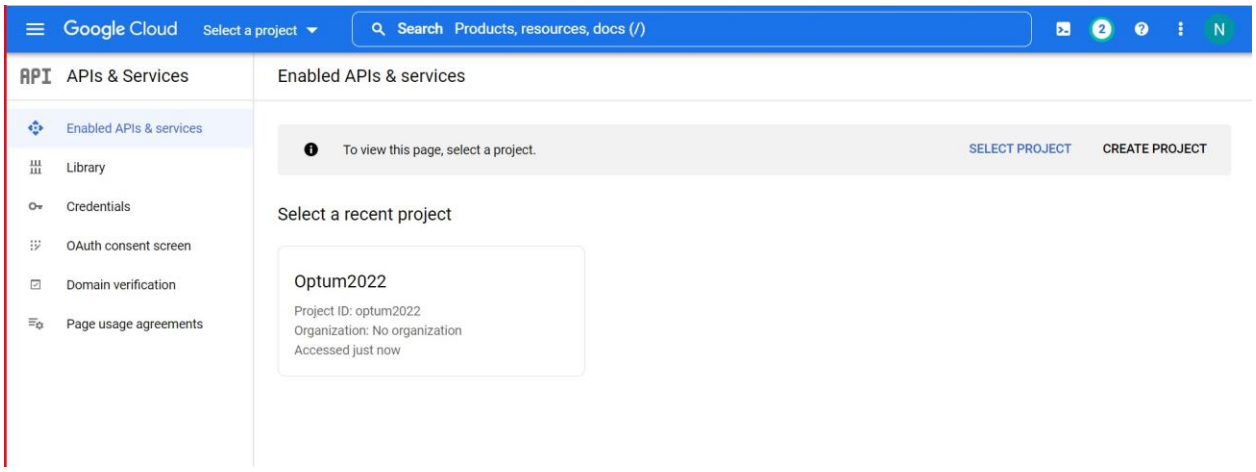
Created a google cloud console account at Google Cloud console with the following credentials:

Username: noreplyuhg@optum.com
Password: Optum2022!

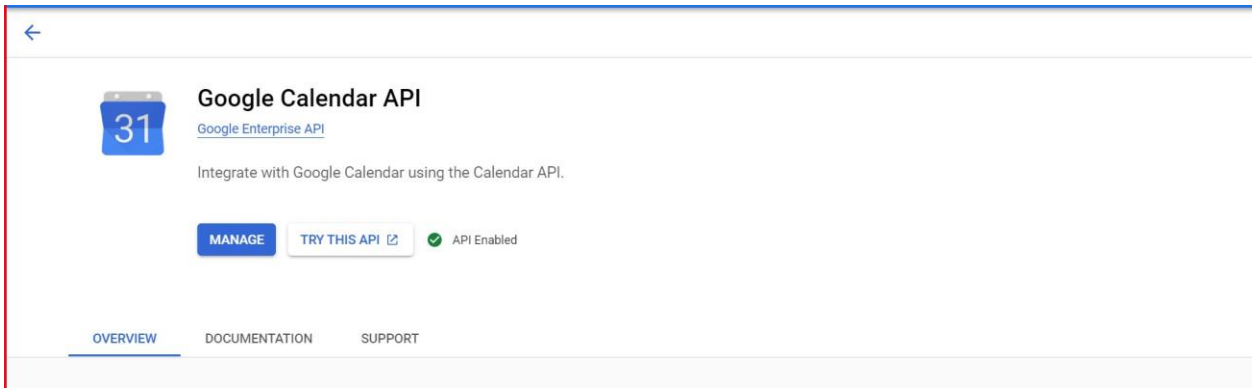
STEP 2: Create a new project by selecting **Create project**.



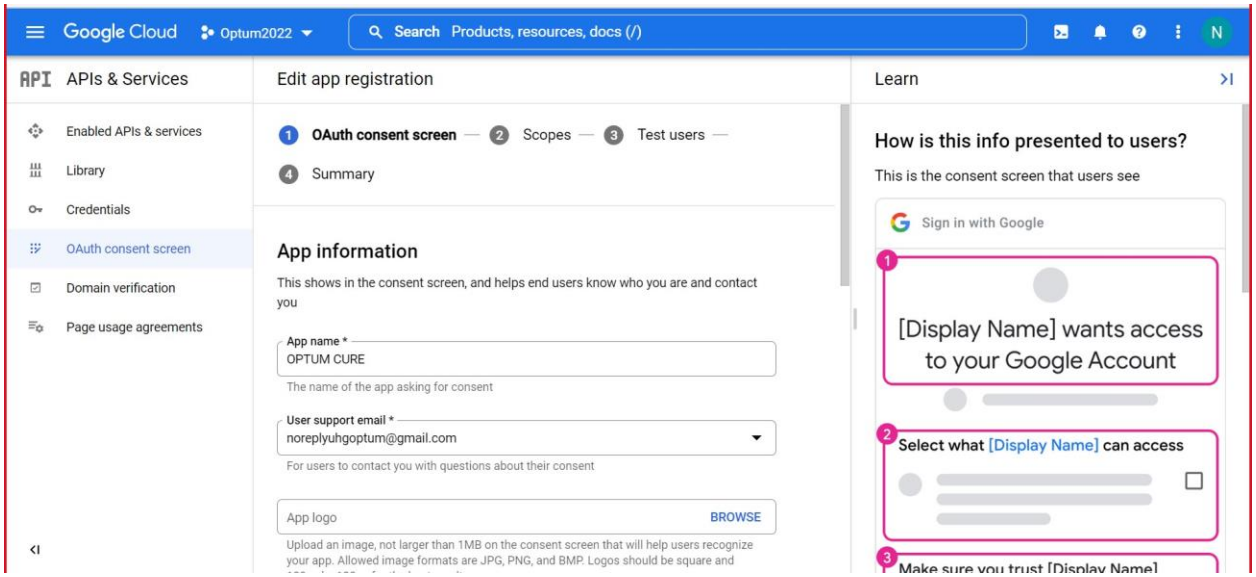


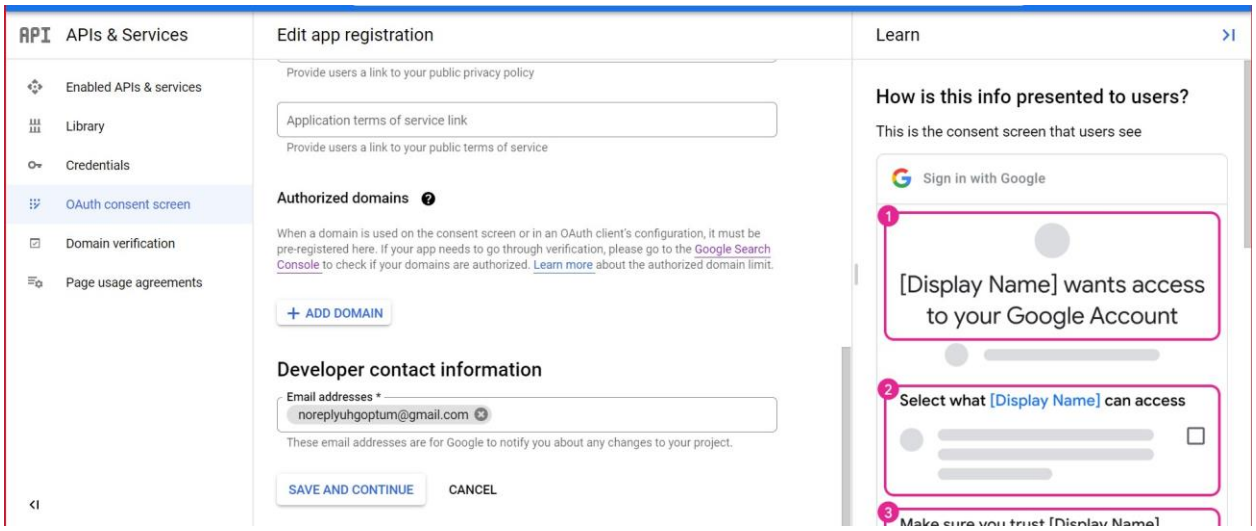


STEP 3: Go to **Enabled API's and Services** and enable the **Google calendar API**.

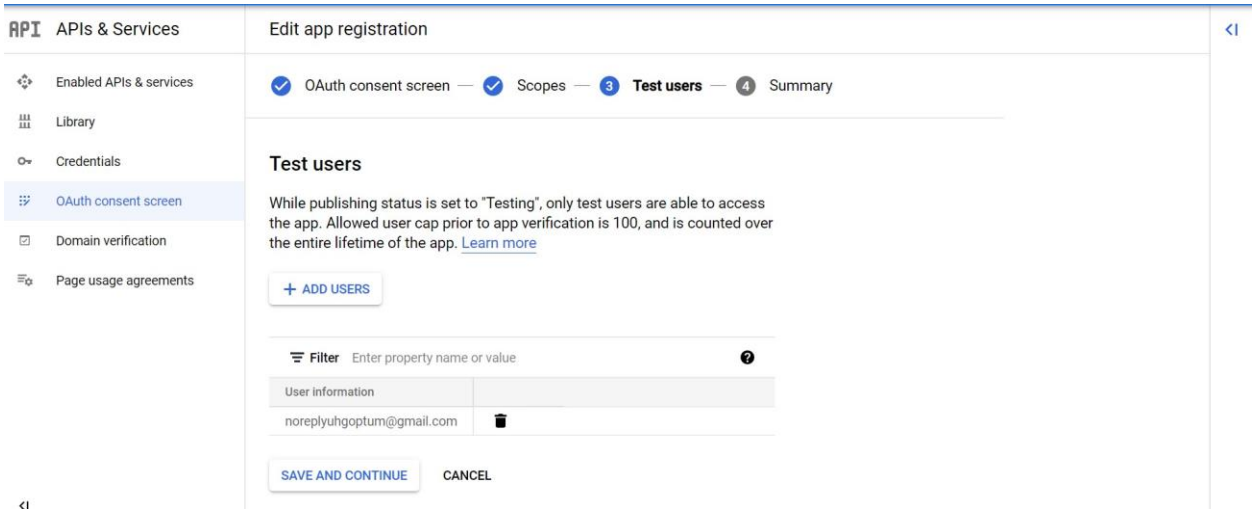


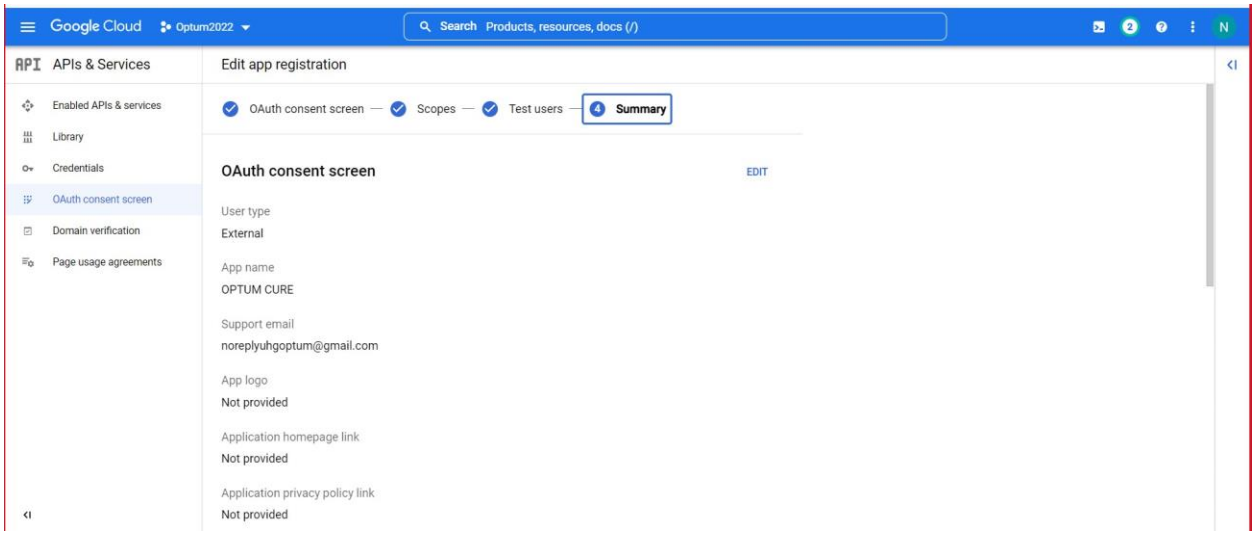
STEP 4: Go to **Oauth Consent screen** and fill the required app information.

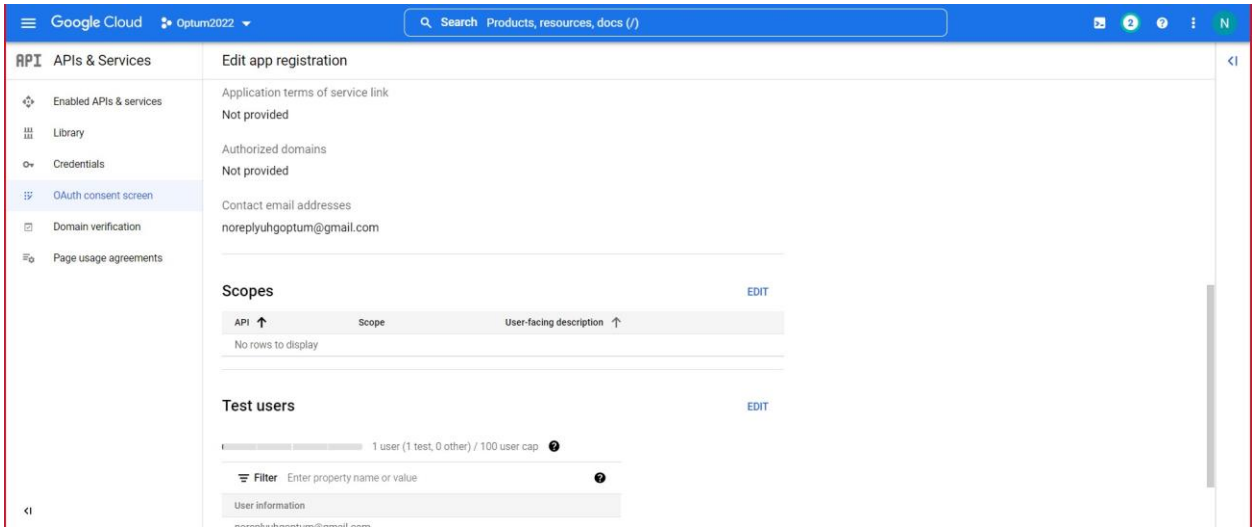




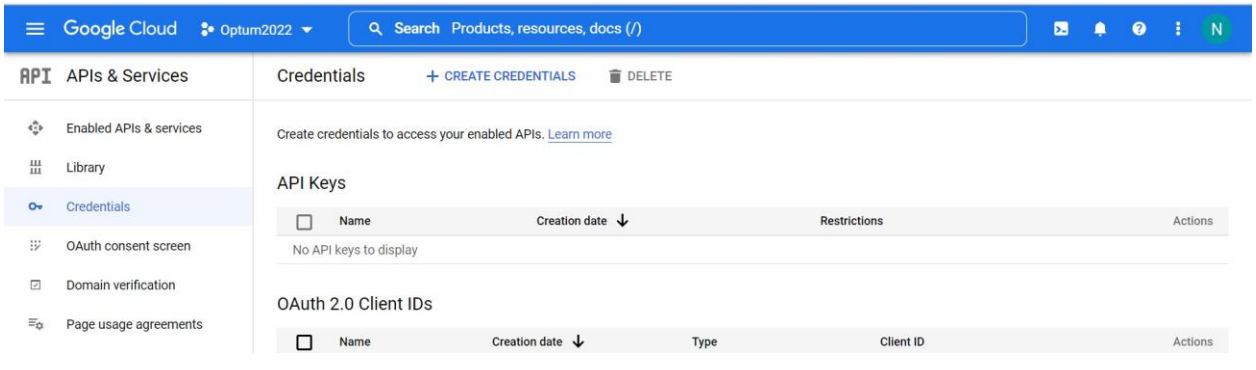
Step 5:Click on save and continue and you will be able to see the **Summary**.

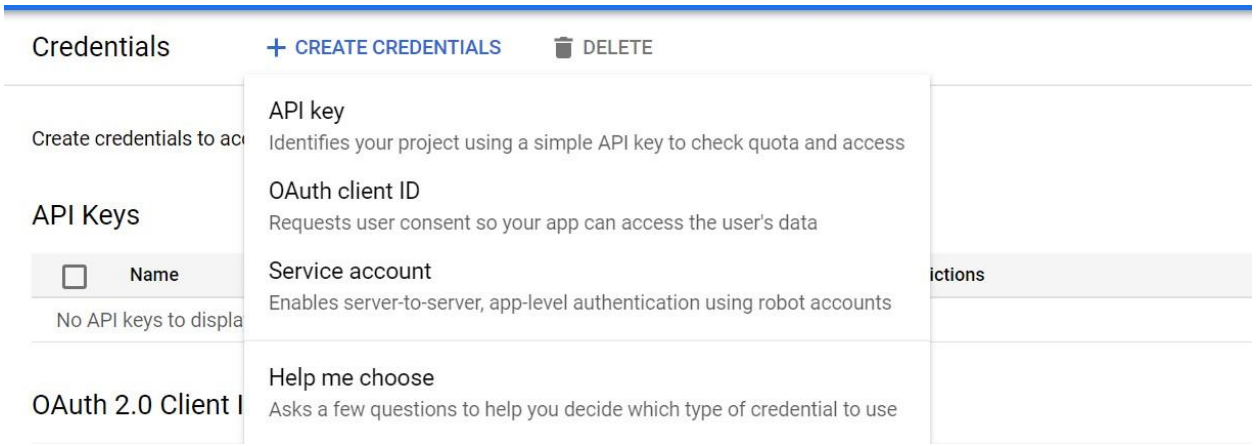


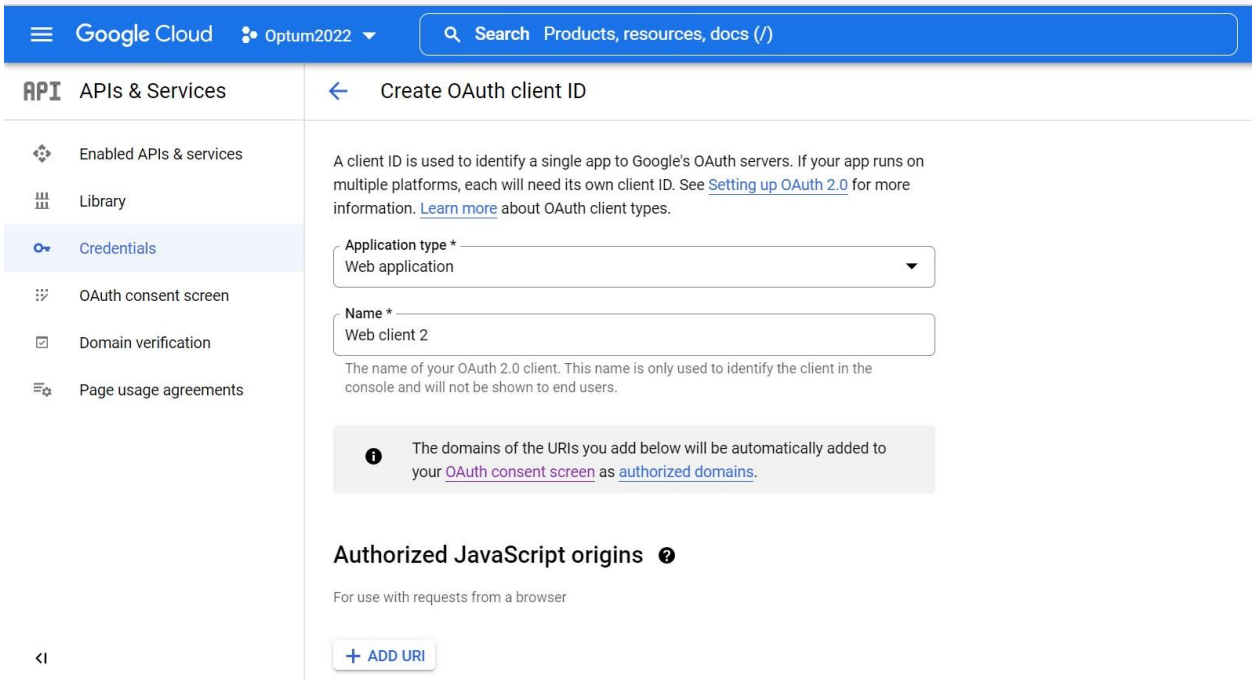


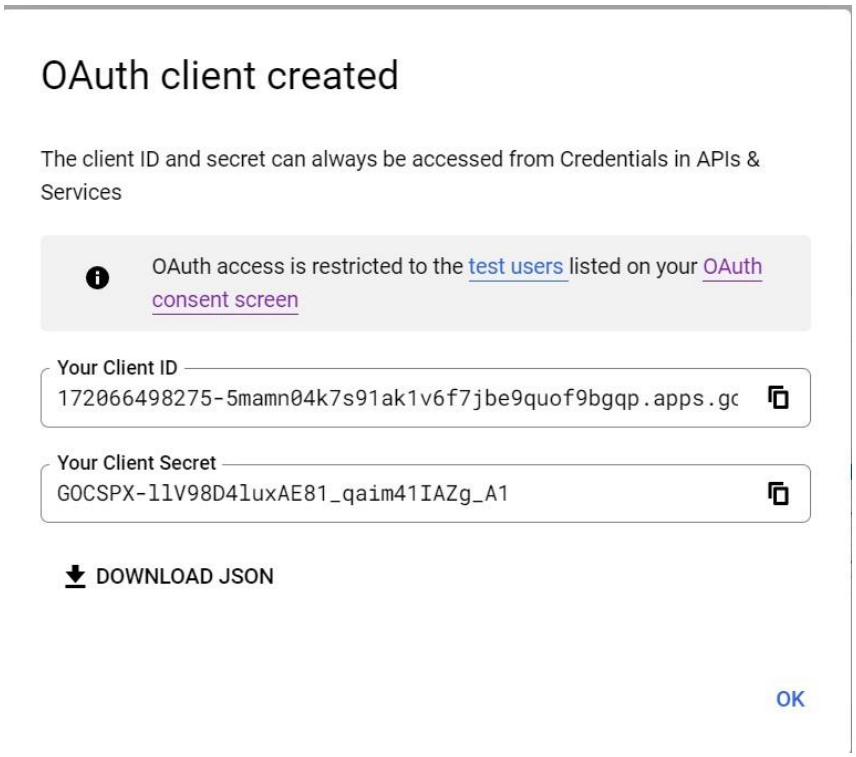


Step 6: Go to **Credentials** and click on **create credentials** and choose **OAuth Client ID.**





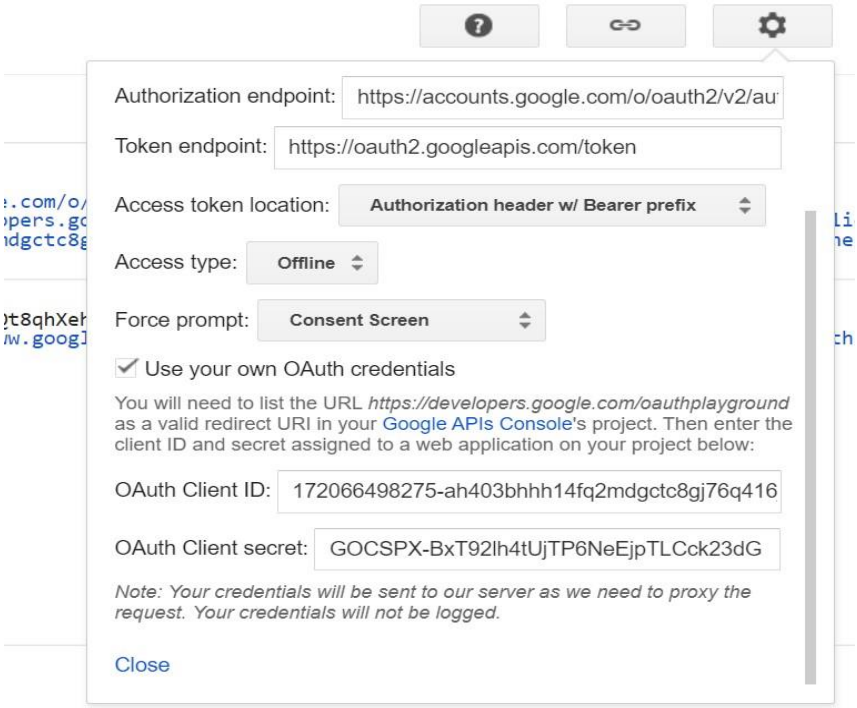




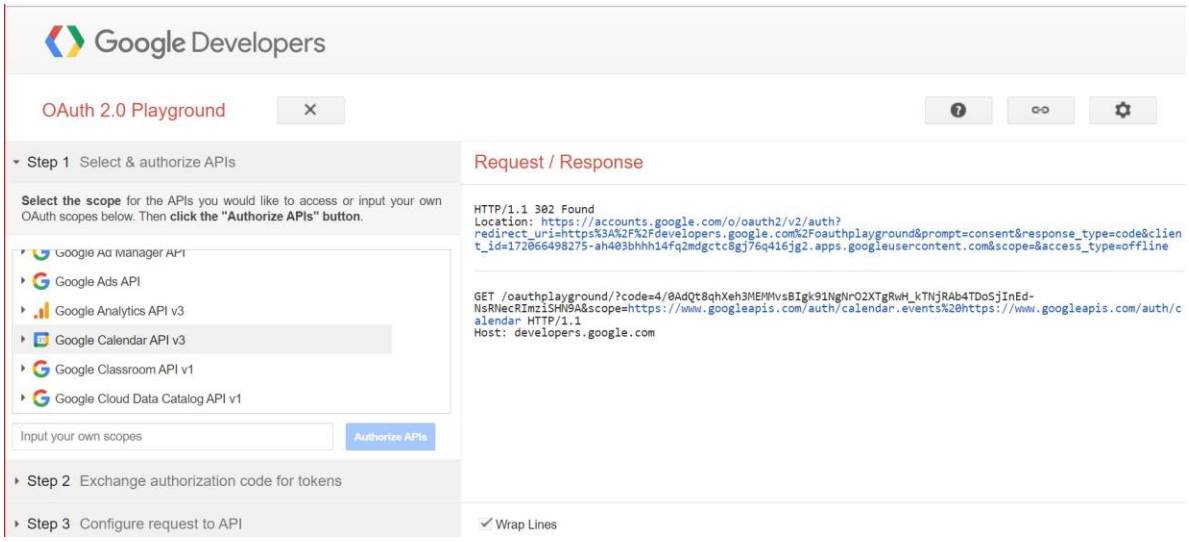
Step 7:Download the **json file** by clicking on **Download Json** which has client secret and client id.

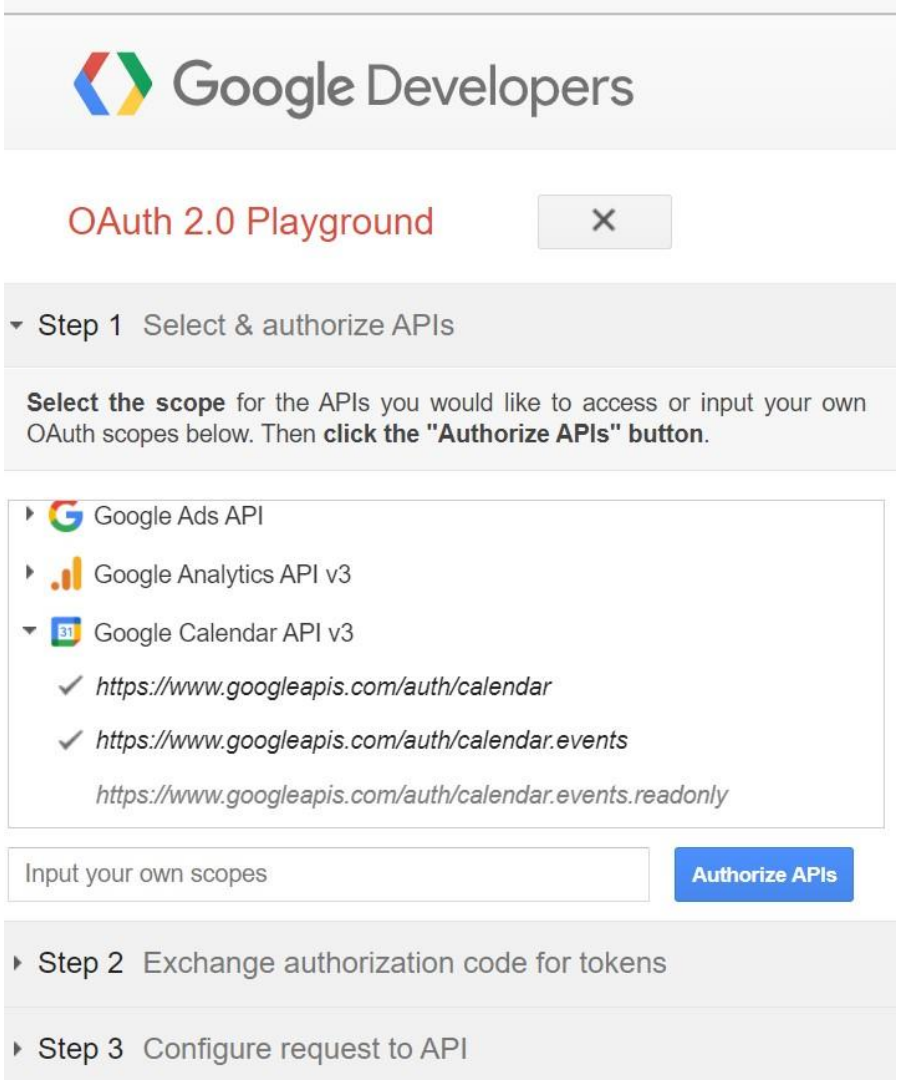
Step 8:Go to [OAuth 2.0 Playground (google.com)]

Step 9:Fill the required Client secret and Client ID.


Authorization endpoint: https://accounts.google.com/o/oauth2/v2/au
Token endpoint: https://oauth2.googleapis.com/token
Access token location: Authorization header w/ Bearer prefix
Access type: Offline
Force prompt: Consent Screen
Use your own OAuth credentials
You will need to list the URL https://developers.google.com/oauthplayground as a valid redirect URI in your Google APIs Console's project. Then enter the client ID and secret assigned to a web application on your project below:
OAuth Client ID: 172066498275-ah403bhhh14fq2mdgctc8gj76q416
OAuth Client secret: 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
Note: Your credentials will be sent to our server as we need to proxy the request. Your credentials will not be logged.
Close

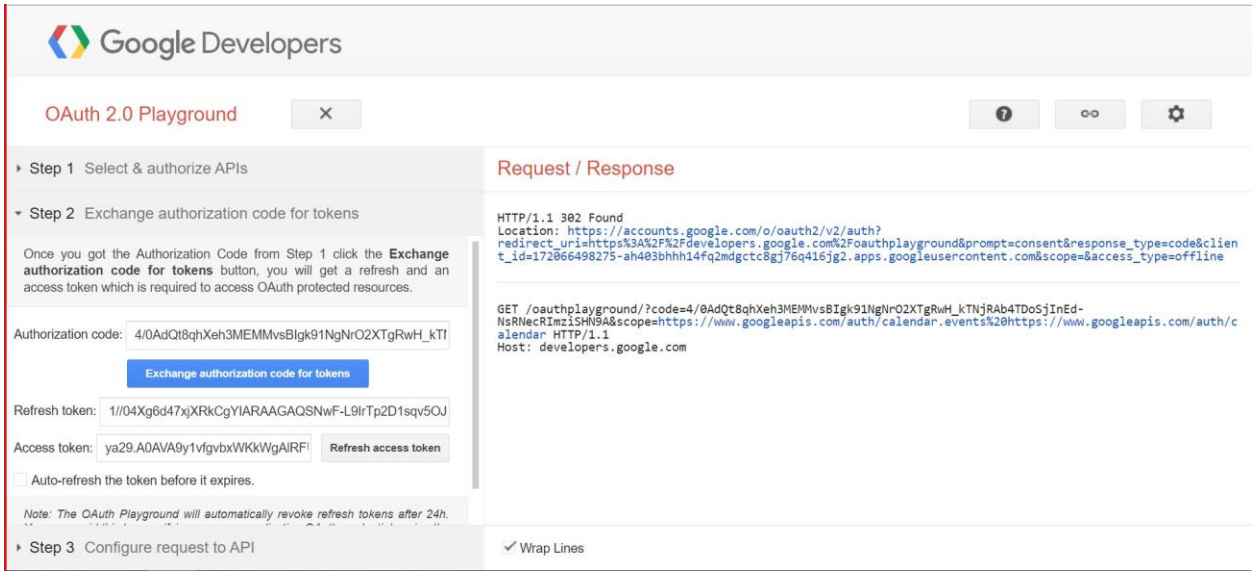
Step 10:From scopes select **Google calendar API v3** and select the Two API's and click on authorize API's.





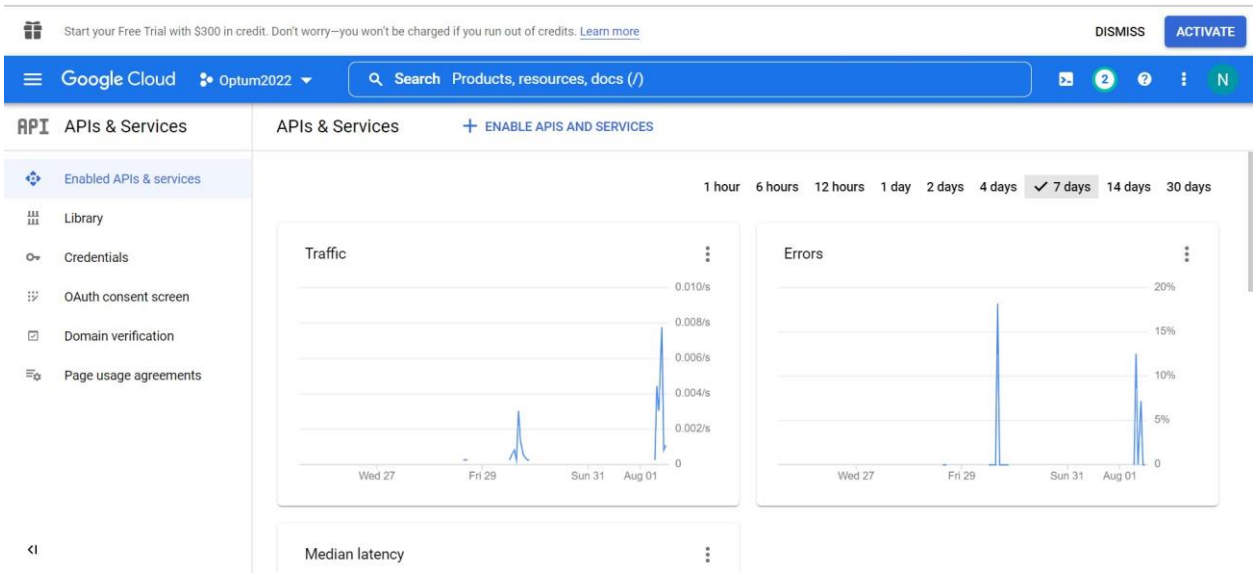
Step 11:Click on authorize API's ie. *https://www.googleapis.com/auth/calendar* and *https://www.googleapis.com/auth/calendar.events*.Then it will ask you to sign in to your google account and after signing in you have to give the access.This will give us the authorisation code.

Step 12:Click on **Exchange authorisation code for tokens**



Step 13:We will get an **access and refresh token** in return.

Step 14:Note it down as it will be used in future.



Step 15:

Install all the dependencies.

Open the cmd and type-

- npm install
- npm init
- npm install googleapis
- npm install requests
- npm install uuid

Step 16:

Booking an appointment:

- Download the json file from the credentials tab.
- Create an index.js file.

STEP 17:

- For the gmail integration, set the parameters in the 'calendar.events.insert' as follows:

1. sendUpdates='all'
2. sendNotifications='true'

Teleconferencing APP Integration

1. Open the command prompt and write npm install uuid.
2. Since we have already installed the uuid package now we just need to import the UUID package and call the method related to the UUID version we want to use.
3. The uuid module generates unique random IDs.
4. Here we are using uuid v1.
5.

```
const uuidv1 = require('uuidv1');
```

6. After retrieving the GUID,concatenate it with the teleconferencing URL and we will get our meeting link.
7.

```
const GUID = uuidv1();
url1=`https://optum-videocallapp.azurewebsites.net/?groupId=${GUID}`
console.log(url1)
```

8. Here url1 is our meeting link.